	SVKM's NMIMS **Mukesh Patel School of Technology Management & Engineering /** **School of Technology Management & Engineering**	
B. Tech CE	**Lab and Workbook**	**Academic Year- 2026-27**
Year:-Third	**Subject:-** Drone Technology	**Semester: - Fifth**

Experiment: 3

PART A

(PART A: TO BE REFERRED BY STUDENTS)

Aim:

a. To understand the interface of Mission Planner.
b. To Create and execute autonomous missions.

Pre-requisite: Desktop with Windows OS, Mission planner software

Learning Outcomes: After completing this experiment, students should be –

1. **Comfortable with using Mission Planner for configuring drones, planning missions, and executing autonomous flights safely and efficiently.**

Theory:

1. Introduction:

- Mission Planner is a ground control station software that provides various tools for drone configuration, mission planning, and monitoring. It supports multiple autopilots, especially for ArduPilot-based drones.

2. Installation and Initial Setup

Step 1: Installing Mission Planner

- Download and install Mission Planner from ArduPilot website.

3. Mission Planning

Step 1: Understanding the Interface

- Explore the map view and mission options on the "Flight Plan" tab.
- Learn how to add waypoints, takeoff commands, landing.

Step 2: Creating a Mission

- Select the "Flight Plan" tab.
- Click on the map to add waypoints, specifying altitude, speed, and actions.
- Add other commands such as:
  - **Waypoint**: For navigation.
  - **RTL (Return to Launch)**: To bring the drone back to its starting point.

Step 3: Uploading the Mission

- Once the mission is planned, click **Write WPs** to save the mission
- Review the mission

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PART B

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Date of Experiment: 15/10/24	**Date of Submission:** 15/10/24

Task to be performed

Task 1: Creating and Executing an Autonomous Mission Using Mission Planner.

Task 1: Creating and Executing an Autonomous Mission Using Mission Planner

Procedure followed

- Installed Mission Planner (Windows) from the ArduPilot website, started it and selected the connection type (COM port / UDP for simulation) with the proper baud rate (115200 for USB, 57600 for telemetry), then clicked **Connect**.
- Observed the **Flight Data** screen: artificial horizon (HUD), map, mode, battery, GPS status and telemetry readouts.
- Opened the **Flight Plan** tab and right-clicked the map at the home location → **Set Home Here**.
- Added a **Takeoff** command with an altitude, then clicked on the map to add waypoints; set altitude and delay for each, and chose the default waypoint altitude and speed.
- Added **RTL (Return to Launch)** as the last command so the drone returns and lands at home.
- Clicked **Write WPs** to upload the mission to the flight controller and **Read WPs** to verify it.
- Armed the drone, took off in a safe area (or in SITL simulation), switched the mode to **AUTO** and watched the drone follow the path and finally return to home with RTL.

Mission created (sample):

No.	Command	Altitude (m)	Parameters / Remark
1	TAKEOFF	20	Climb vertically to 20 m
2	WAYPOINT	20	Delay 0 s – first corner of the rectangle
3	WAYPOINT	20	Second corner
4	WAYPOINT	20	Third corner

5	WAYPOINT	20	Fourth corner (path completes a rectangle)
6	RETURN_TO_LAUNCH	–	Return to home and land

Mission Planner interface elements used:

Tab / Control	**Use**
Flight Data	Live telemetry, HUD, arm/disarm, change flight mode
Flight Plan	Add/edit waypoints, Set Home, Read/Write WPs
Initial Setup	Frame type, calibration, radio, flight modes
Config/Tuning	Parameters and PID tuning
Simulation	Run SITL for testing missions without hardware

[Paste screenshots here: (a) Mission Planner main screen, (b) Flight Plan tab with waypoints, (c) Write WPs success, (d) Flight Data tab in AUTO mode]

Observations and Learning:

- Understood the Mission Planner interface and its functionalities for drone configuration.
- Learned how to add waypoints, set altitude, speed, and other commands for mission execution.
- Successfully created and uploaded an autonomous mission using the "Flight Plan" tab.
- Observed how the drone follows the planned path and executes commands like Waypoint and RTL.
- Identified the importance of pre-flight checks and mission validation before execution.

Conclusion:

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Mission Planner is an effective tool for configuring and executing autonomous drone missions. It simplifies mission planning with an intuitive interface, allowing precise control over flight paths. By using features like waypoint navigation and RTL, users can ensure safe and efficient drone operations.